CICFlowMeter CSV is the standard input for IDS classification

Prior experiments: XGB · TabPFN · SOTA · proposed models → CICFlowMeter CSV vectors

Most IDS tabular-classification studies also train and evaluate on the supplied flow-statistics CSV

Input to prior classifiers

One CSV row

[duration, packets, length, IAT, flags, ...]
→ numeric vector + label

Each row treated as an independent sample

CIC-IDS2017: How PCAP becomes CSV

Testbed	PCAP	CICFlowMeter	CSV
benign + attack	raw packets in time order	flows + fixed statistics	vector + label

Examples of fixed flow statistics in the CSV

Flow duration	Fwd. packets	Packet interval
first to last packet	packets sent	interval statistics

A summary of predefined flow statistics, not the packet sequence itself

Two issues in CSV-based IDS classification

Issue 1 Early saturation in some attacks

Head attacks such as DDoS and Bot reach F1 in the 90% range with n = 2 to 20

→ a high data ceiling before adding labels

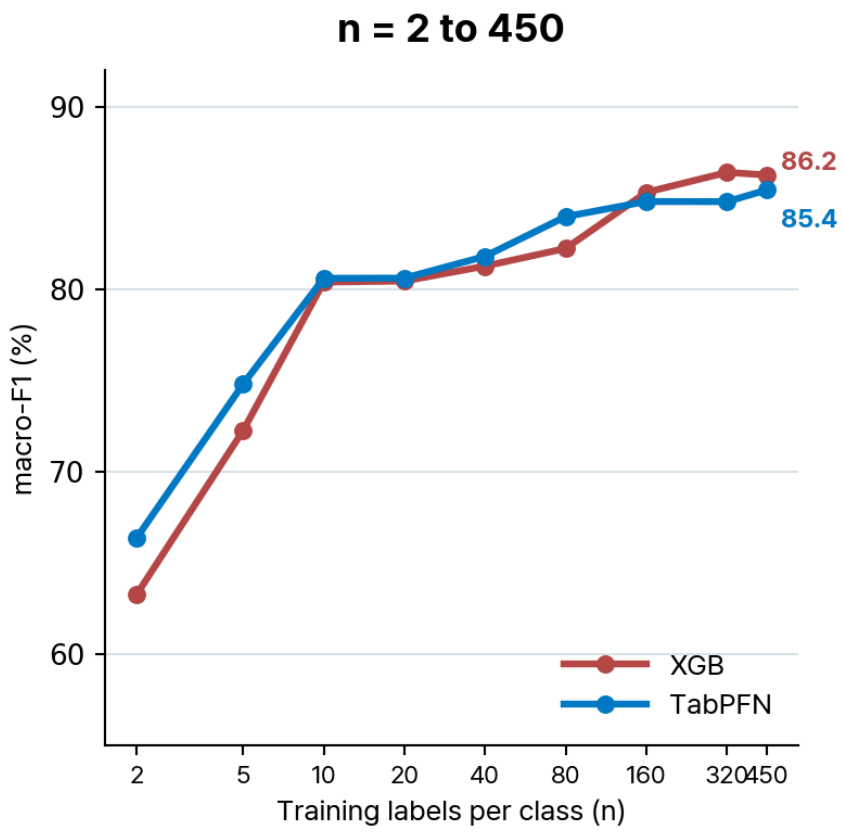
Issue 2 Identical features, different labels

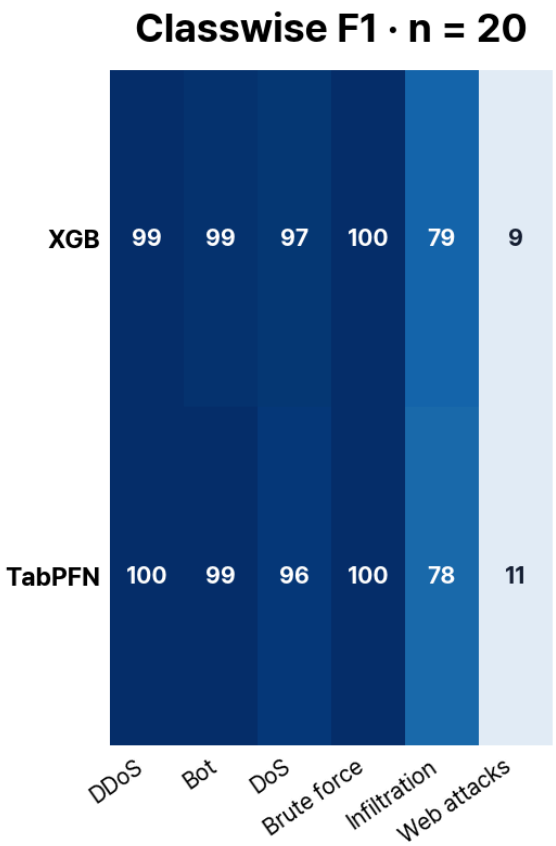
Rows with dozens of identical flow statistics can still have different recorded labels

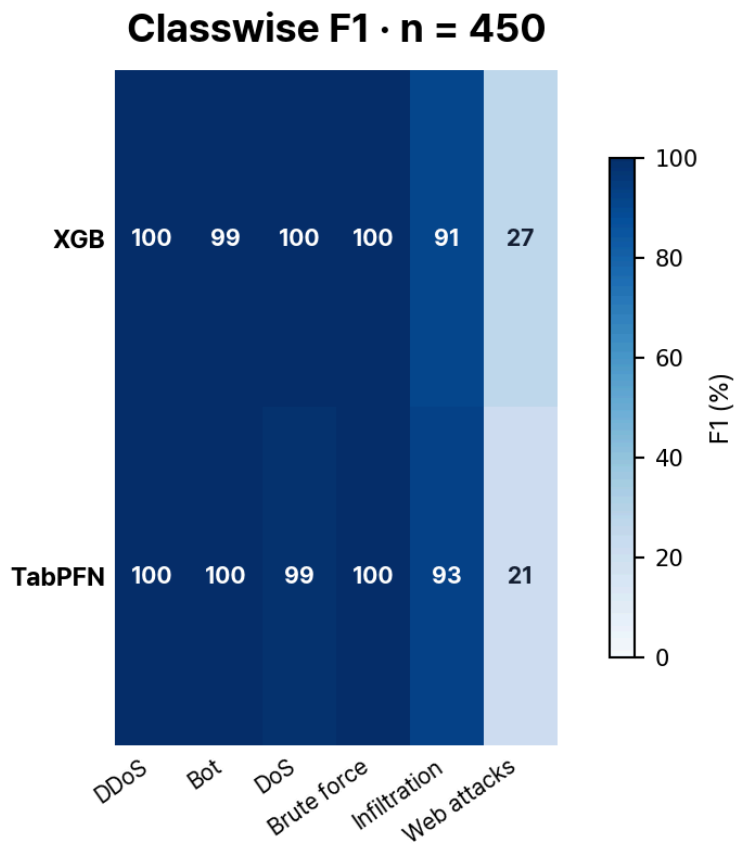
→ one deterministic classifier cannot fit both

Some head attacks already have high F1 with few labels

Macro F1 still changes beyond n = 20, but DDoS and Bot approach high F1 with few labels







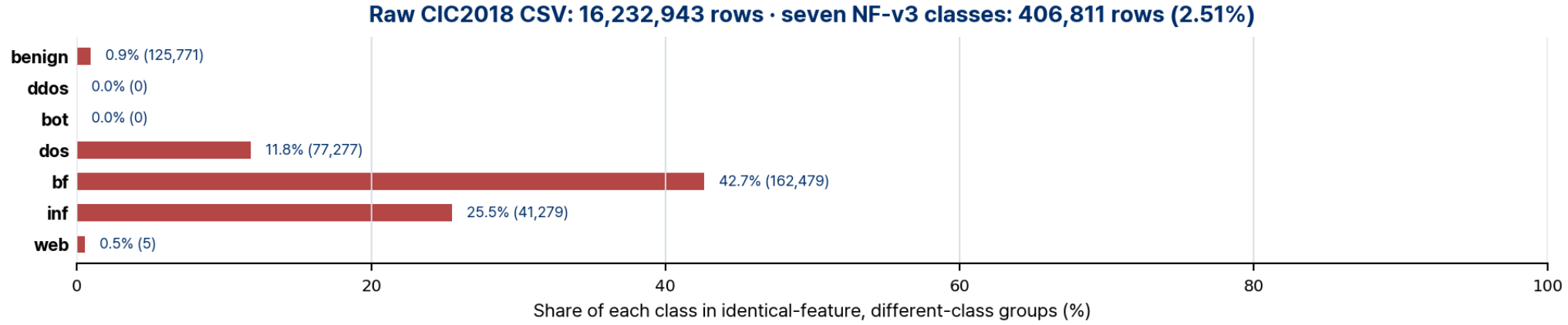
EXP87 · attack-only training and evaluation · benign excluded · 5 seeds · fixed conflict-free test · n = 450 is the balanced-label-budget ceiling

Identical features, different classes in CIC2018 CSV

Exact agreement on 78 common CICFlowMeter statistics, excluding timestamp, Flow ID, source IP/port, destination IP, and label

Dst Port	Protocol	Flow Duration	Tot Fwd Pkts	Tot Bwd Pkts	... 69 fields ...	TotLen Bwd Pkts	Flow Byts/s	Flow Pkts/s	Idle Min	NF-v3 class
21	6	2	1	1	... 69 fields ...	0	0	1000000	0	**dos**
21	6	2	1	1	... 69 fields ...	0	0	1000000	0	**bf**

The two rows above: bf · dos · all other 69 fields are also identical



Audit of all 16,232,943 raw CSV rows · aggregated to seven NF-v3 classes · bar endpoint: different-class rows

A PCAP-based CIC2018 representation pipeline

Generate representations from packet sequences, then recheck both issues under the same protocol

PCAP + labels	Flow linkage	Encoder	Projection	New vectors
packet, time, label evidence	traceable traffic and labels	packet-sequence embedding	fit PCA on the training split only	XGB + TabPFN same evaluation

Representation audit	Classification evaluation
identical embeddings with different labels and the PCAP evidence for each label	baselines, few-shot curves, and classwise F1 to test whether early head saturation remains

Candidate encoders netFound (arXiv, 2026) · ET-BERT (WWW, 2022) · NetMamba (ICNP, 2024)

CICFlowMeter uses specified statistics; encoders learn from packet order, timing, and headers

Summary

1 Existing CSV input

XGB · TabPFN · SOTA
CICFlowMeter flow statistics
row-level inputs

2 Observed data issues

early head saturation
identical features, different classes

average score alone
does not establish difficulty

3 Next validation

new vectors from PCAP

repeat audits and baselines
under the same protocol

Goal Test whether new representations reduce the data ceiling and feature–label conflicts

If reduced, repeat TabPFN and XGB comparisons on the new vectors; if retained, examine PCAP label linkage further